

ECON 164: Theory of Economic Growth

Week 5: AK Models

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 - **Next few lectures**: models of “endogenous growth”

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- anchor to Solow, but everything we say applies to Ramsey-Cass-Koopmans too

The AK model: A limiting case of the Solow model

What happens in the **Solow model**...

(with *constant* Hicks-neutral productivity)

$$Y_t = AK_t^\alpha (L_t)^{1-\alpha}, \quad \dot{K}_t = sY_t - \delta K_t$$

...if we set $\alpha = 1$?

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...if we set $\alpha = 1$? The model becomes...

$$Y_t = AK_t, \quad \dot{K}_t = sY_t - \delta K_t$$

...so the growth rate of capital becomes

$$g_K = \frac{\dot{K}_t}{K_t} = sA - \delta$$

(Problem Set #2, Question 1B)

Dynamics when $\alpha = 1$: AK model generates sustained growth thru K

Growth rate of output per capita:

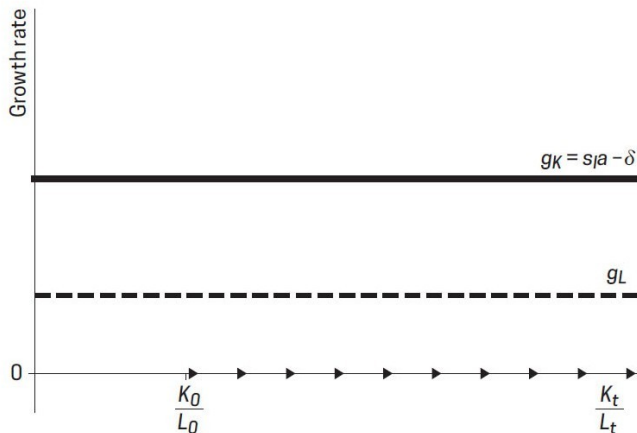
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Figure 11.1 Dynamics of the AK Model



NOTE: The growth rate of capital, g_K , does not depend on the ratio K_t/L_t , so the two curves never cross and there is permanent growth in K_t/L_t over time.

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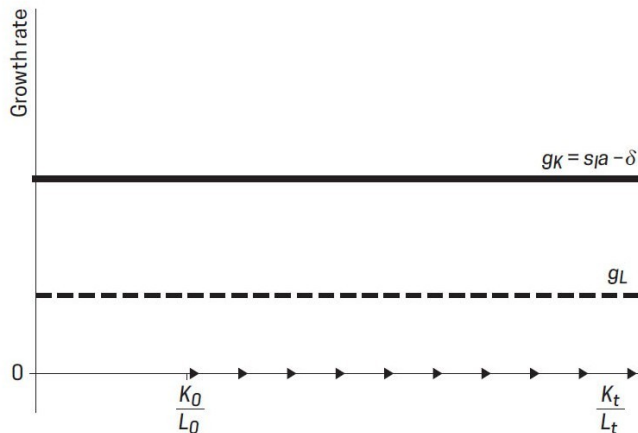
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Some observations:

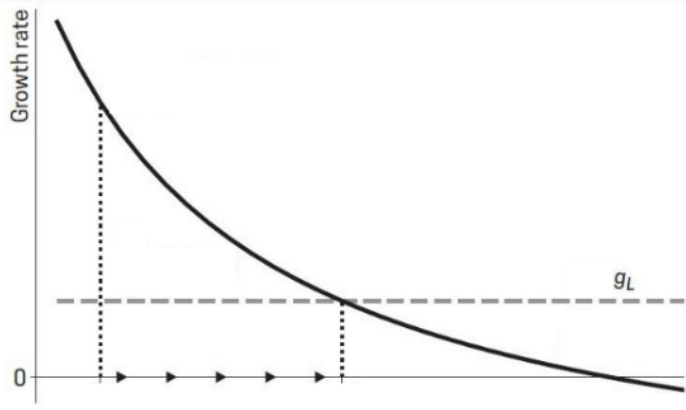
- g_y is **independent** of k_t
- g_y is **positive** and **persistent** even w/ $g_A = 0$
- g_y is affected **permanently** by a change in policy (s)

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Intuition: It's all about (avoiding) diminishing returns...



For $\alpha < 1$...

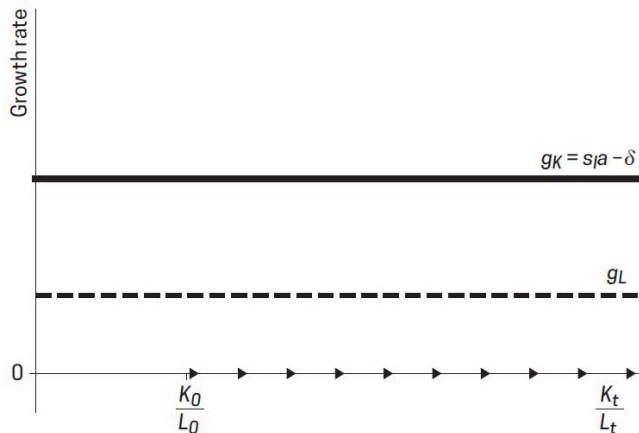
$$\dot{k} = sAk^\alpha - (\delta + g_L)k$$

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$$g_y = \alpha(sAk^{\alpha-1} - \delta - g_L)$$

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For $\alpha < 1$...

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$$g_y = \alpha(sAk^{\alpha-1} - \delta - g_L)$$

... but with $\alpha = 1$:

$$\dot{k} = sAk - (\delta + g_L)k$$

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$$g_y = sA - \delta - g_L$$

... somewhere in the model

In the **Solow model** *with* productivity growth...

→ *sustained growth thru A_t*

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In **general**, there must be some variable X_t governed by a **linear differential equation**

$$\dot{X}_t = g_X X_t$$

(*why isn't $\dot{L}_t = g_L L_t$ enough then?*)

- Recall the **production function** with human capital (but *constant* Hicks-neutral A)

$$Y_t = AK_t^\alpha (u h_t L_t)^{1-\alpha}$$

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- Accumulating **physical capital** happens as it does in Solow...

$$\dot{K}_t = sY_t - \delta K_t$$

- ... but now we explicitly accumulate **human capital**, too:

$$\dot{h}_t = (1 - u)h_t$$

The Uzawa-Lucas model

On the one hand, this is a cousin of the **AK model** → **same long-run dynamics**

- $\dot{h}_t$ is linear in h_t
- $\dot{K}_t$ is linear in $\{K_t, h_t\}$ because $Y_t = F(K_t, h_t, L_t)$ is **CRS** in that tuple

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On the other, it's still a cousin of the **Solow model** → **same transitional dynamics**

- h_t acts like **labor-augmenting** productivity
- balanced growth path (BGP) requires **constant** $\hat{k}_t = \frac{K_t}{h_t L_t} \dots$
- ...so both s and u affect the level, but only u affects the growth rate of y_t

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- This maintains assumption of **perfect competition** → **vs. next week**

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1. growth vs. level effects
2. scale effects
3. what do firms actually do?

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[advanced reading: [McGrattan \(1998\)](#)]

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- So larger countries should grow **faster**... but this is **not** what we see (SGP vs. IND)
- A large literature worked out models to eliminate these **scale effects**...
 - [Segerstrom \(1998\)](#); [Young \(1998\)](#); [Howitt \(1999\)](#); [Jones \(1999\)](#); [Peretto \(2018\)](#)
 - Uzawa-Lucas avoids it b/c what matters is **average**, not **total**, human capital
- ...but maybe we just need to think harder about the right **spatial scale**:
 - maybe it's the **whole world** in the very long run ([Kremer, 1993](#))
 - maybe it's just **local labor markets**

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That's what we cover the next two weeks!

What *is* endogenous growth?

- **our running definition:** long-run growth determined by modeled choices (Slide 3)
- **the 1980–90s answer:** permanent changes in policies have **permanent effects** on an economy's long-run growth rate
- **the modern answer:** the endogenous outcome, **permanent** or **transitory**, of an economy in which profit-seeking individuals who are allowed to earn rents on the fruits of their labors search for newer and better ideas ([Jones and Vollrath, 2024, p.241](#))

when only transitory, prefer to call this ***semi-endogenous***

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- but, at best, innovation is **accidental** $\rightarrow$ **next two weeks**

References

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